Advanced Macroeconomics

I. Foundations of Dynamic Macroeconomic Modeling
   1. Introduction
   2. History of Macroeconomics
   3. Static General Equilibrium Models
   4. Saving and Investment in a Two-Period Model

II. Long-run Economic Growth

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Does Higher Productivity Mean More Employment?

A Productivity Improvement

Suppose a technological improvement allows firms to produce more output with the same amount of labor.

Productivity increases: $A \uparrow$

- What happens to equilibrium output, consumption, and real wages?
- Does higher productivity lead to more or less employment?
- Which economic mechanisms determine your predictions?

3. Static General Equilibrium Models

Models, Variables, and Functions
General Equilibrium in a One-Period Model with Fixed Endowment
General Equilibrium in a One-Period Model with Endogenous Labor
Summary and Literature

Building a Macroeconomic Model

- Purpose: Translate an economic question into a system of equations
- Building blocks:
  - Behavioral relations: household and firm decisions
  - Technology and accounting: production and resource constraints
  - Equilibrium conditions: supply equals demand
- Variables: Endogenous variables are determined by the model; exogenous variables are taken as given
- Parameters: Given characteristics of preferences and technology

How does a change in productivity affect the equilibrium of our economy?

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A One-Period Competitive Economy

- Households: A representative household maximizes utility from consumption $u(c)$ and supplies capital k and labor n
- Firms: A representative competitive firm maximizes profits and produces output using k and n:

$$y = AF(k, n)$$

- Endowments: The household has one unit of capital and one unit of time:

$$0 \leq k \leq 1, \qquad 0 \leq n \leq 1$$

- One-period setting: No investment; the capital stock is given as an endowment and can be consumed at the end of the period
- Household budget constraint:

$$c \leq (1 + r)k + wn$$

where w is the real wage and r is the net return on capital

Household Optimization in the One-Period Model

- Utility maximization

$$\max_{c,k,n\geq 0} u(c) \quad \text{s.t.} \quad \begin{cases} c \leq (1+r)k + wn, \\ k \leq 1, \quad n \leq 1 \end{cases}$$

- Lagrangian

$$\mathcal{L} = u(c) - \lambda_1[c - (1+r)k - wn] - \lambda_2(k-1) - \lambda_3(n-1)$$

- First-order conditions

$$\begin{aligned} u'(c) &= \lambda_1 > 0, \\ \lambda_2 &= (1+r)\lambda_1 > 0, \\ \lambda_3 &= w\lambda_1 > 0 \end{aligned}$$

  assuming $u'(c) > 0$, $1 + r > 0$, and $w > 0$.

- Optimal allocation All constraints are binding:

$$\boxed{k = 1 \qquad n = 1 \qquad c = 1 + r + w}$$

Production and Profit Maximization

- Cobb-Douglas production function

$$y = Ak^{\alpha}n^{1-\alpha}, \qquad A > 0, \quad 0 < \alpha < 1$$

- Profit maximization

$$\max_{k,n} \; D = Ak^{\alpha}n^{1-\alpha} - rk - wn$$

- First-order conditions

$$r = \frac{\partial y}{\partial k} = \alpha A k^{\alpha-1} n^{1-\alpha},$$

$$w = \frac{\partial y}{\partial n} = (1-\alpha) A k^{\alpha} n^{-\alpha}$$

- Constant returns to scale: Factor income exhausts output

$$rk + wn = y \qquad \Rightarrow \qquad D = 0$$

General Equilibrium in the One-Period Model

- Factor supplies: The household supplies its entire endowment:

$$k = 1 \qquad n = 1$$

- Equilibrium output and factor prices:

$$y = A \qquad r = \alpha A \qquad w = (1 - \alpha)A$$

- Equilibrium consumption

$$c = (1 + r)k + wn = 1 + A$$

- Factor income shares

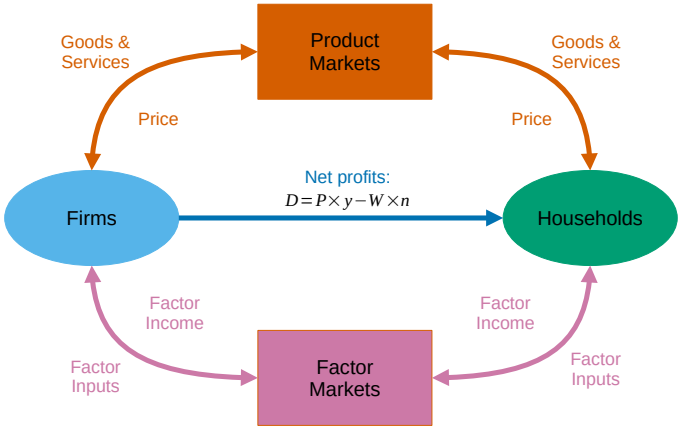
$$\underbrace{\frac{rk}{y} = \alpha}_{\text{Capital share}} \qquad \underbrace{\frac{wn}{y} = 1 - \alpha}_{\text{Labor share}}$$

- Back to our productivity puzzle: Higher productivity increases output, consumption, and factor prices – but employment remains fixed by assumption

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Walrasian Equilibrium



Walrasian Equilibrium with Endogenous Labor Supply

- Households: Choose consumption c and labor supply n to maximize utility $u(c, n)$, subject to

$$Pc \leq Wn + D$$

  where D denotes distributed firm profits

- Firms: Produce output using labor

$$y = An^{1-\alpha}, \qquad 0 < \alpha < 1,$$

  and maximize profits $D = Py - Wn$

- Market clearing: Goods supply equals demand ($y = c$); labor supply equals labor demand
- General equilibrium: Determine the four endogenous variables (w, n, y, c), where $w = W/P$ is the real wage; nominal wages and prices are not separately determined

Labor supply is now endogenous. Unlike the previous benchmark, this model does not include a capital endowment that can be consumed.

Walrasian Equilibrium: Household Optimization

- Utility maximization

$$\max_{c,n} u(c,n) \qquad \text{s.t.} \qquad Pc = Wn + D$$

with

$$u(c,n) = \frac{c^{1-\sigma}-1}{1-\sigma} - \frac{n^{1+\varphi}}{1+\varphi} \qquad \lim_{\sigma \to 1} \frac{c^{1-\sigma}-1}{1-\sigma} = \ln c$$

- Lagrangian

$$\mathcal{L} = \frac{c^{1-\sigma}-1}{1-\sigma} - \frac{n^{1+\varphi}}{1+\varphi} + \lambda(Wn + D - Pc)$$

- First-order conditions

$$\frac{\partial \mathcal{L}}{\partial c} = c^{-\sigma} - \lambda P = 0 \qquad \text{and} \qquad \frac{\partial \mathcal{L}}{\partial n} = -n^{\varphi} + \lambda W = 0$$

- Labor supply condition

$$\boxed{w \equiv \frac{W}{P} = c^{\sigma} n^{\varphi}}$$

Walrasian Equilibrium: Firm Optimization

- Profit maximization

$$\max_{y,n} D = Py - Wn \qquad \text{s.t.} \qquad y = An^{1-\alpha}$$

- Lagrangian

$$\mathcal{L} = Py - Wn - \kappa(y - An^{1-\alpha})$$

- First-order conditions

$$\frac{\partial \mathcal{L}}{\partial y} = P - \kappa = 0 \qquad \text{and} \qquad \frac{\partial \mathcal{L}}{\partial n} = -W + \kappa A(1-\alpha)n^{-\alpha} = 0$$

- Labor demand condition

$$\boxed{w \equiv \frac{W}{P} = A(1-\alpha)n^{-\alpha}}$$

- Firm profits (distributed to the household)

$$\frac{D}{P} = y - wn = \alpha y > 0$$

Walrasian Equilibrium: Equilibrium Conditions

- Four endogenous variables Real wage w, employment n, output y, and consumption c
- Labor supply (households)

$$w = c^{\sigma} n^{\varphi}$$

- Labor demand (firms)

$$w = A(1-\alpha)n^{-\alpha}$$

- Goods market clearing

$$y = c$$

- Production function

$$y = An^{1-\alpha}$$

- General equilibrium: Four equations determine four endogenous variables; the nominal wage W and price level P are not separately determined

Walrasian Equilibrium: Linear System in Logs

Taking natural logarithms of the equilibrium conditions yields

- Labor supply

$$\ln w = \sigma \ln c + \varphi \ln n$$

- Labor demand

$$\ln w = \ln A + \ln(1 - \alpha) - \alpha \ln n$$

- Goods market clearing

$$\ln y = \ln c$$

- Production function

$$\ln y = \ln A + (1 - \alpha) \ln n$$

The system is exactly linear in log variables. We can express it as a system of linear equations and solve it analytically or numerically.

Solving Systems of Linear Equations

- Example

$$2x_1 + 3x_2 = 18,$$
$$3x_1 - 4x_2 = -7$$

- Matrix representation

$$B\mathbf{x} = \mathbf{b}, \qquad B = \begin{pmatrix} 2 & 3 \\ 3 & -4 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 18 \\ -7 \end{pmatrix}$$

- Numerical solution in Matlab/Octave

$$\mathbf{x} = B^{-1}\mathbf{b} \qquad \longrightarrow \qquad \texttt{x = B\b;}$$

- Result

$$\boxed{\mathbf{x} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}}$$

Hands-on Exercise 1: Solving Linear Equations

Your Task

Open `linsyssolv.m` in Matlab or Octave.

1. Run the script. Verify that both solution methods give the same result

$$\mathbf{x} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

2. Modify the system. Change the first element of **b** from 18 to 19 and run the script again
3. Compare the results. How do the two elements of the solution vector change?
4. Verify your solution. Enter `B*x1-b` in the command window. What result should you expect?

Numerical Solution of the Walrasian Model

- Linear system in matrix notation

$$\mathbf{Bx} = \mathbf{b}, \qquad \mathbf{x} = \begin{pmatrix} \ln y & \ln c & \ln w & \ln n \end{pmatrix}'$$

- Coefficient matrix and right-hand side

$$\mathbf{B} = \begin{pmatrix} 0 & -\sigma & 1 & -\varphi \\ 0 & 0 & 1 & \alpha \\ 1 & -1 & 0 & 0 \\ 1 & 0 & 0 & -(1-\alpha) \end{pmatrix} \qquad \mathbf{b} = \begin{pmatrix} 0 \\ \ln A + \ln(1-\alpha) \\ 0 \\ \ln A \end{pmatrix}$$

- Numerical solution in Matlab/Octave

$$\mathbf{x} = \mathbf{B}^{-1}\mathbf{b}$$

```
Log_x_vec = B_Mat\b_vec
```

- Recover equilibrium values: Exponentiate the logarithms to obtain (y, c, w, n) in levels

Hands-on Exercise 2: Comparative Statics

Your Task

Open `linwalras.m` in Matlab or Octave; investigate how a productivity increase affects the Walrasian equilibrium

1. Compute the baseline equilibrium. Run the script with $A = 1$ and record the equilibrium values of w, n, y, and c
2. Introduce a productivity shock. Change the productivity parameter in the code

   ```
   A_sca = 1.1
   ```

   Run the script again
3. Compare the equilibria. Which variables increase, decrease, or remain unchanged?
4. Interpret the results. Why does employment respond as it does? Which model assumptions are important?

Solving the Walrasian Model with Dynare

- Dynare: Software for solving macroeconomic models, integrated with Matlab or Octave
  - Model equations are specified in a `.mod` file
  - The same framework can later handle dynamic stochastic models
- Model specification: The file `walras.mod` contains
  - Endogenous and exogenous variables
  - Parameters and their values
  - Four equilibrium equations in the `model(linear)` block
- Computing the equilibrium: The command `steady` solves the static equilibrium system; results are stored in `oo_.steady_state`
- Important: The model is specified in logarithms

$$a = \ln A, \qquad (w, n, y, c)_{\text{Dynare}} = (\ln w, \ln n, \ln y, \ln c)$$

Exponentiate the solution to obtain levels

Hands-on Exercise 3: Solving the Model with Dynare

Your Task

Open `walras.mod` and solve the Walrasian equilibrium using Dynare

1. **Run the model.** Execute in Matlab/Octave:

   ```
   dynare walras.mod
   ```

   Compare the results with `linwalras.m`.

2. **Introduce a productivity shock.** Change the initialization to

   ```
   a = log(1.1);
   ```

   and rerun Dynare

3. **Compare the equilibria.** Do you obtain the same responses of w, n, y, and c as in Exercise 2?

4. **Understand the code.** Why must we exponentiate the results?

Solving the Nonlinear Walrasian Model

- The problem. In general, equilibrium conditions are nonlinear in the endogenous variables; we seek a vector $\mathbf{x}$ such that

$$\mathbf{F}(\mathbf{x}) = \mathbf{0}.$$

- Equilibrium residuals. For $\mathbf{x} = (y, c, w, n)'$:

$$\mathbf{F}(\mathbf{x}) = \begin{pmatrix} w - c^{\sigma} n^{\varphi} \\ w - (1-\alpha) A n^{-\alpha} \\ y - c \\ y - A n^{1-\alpha} \end{pmatrix} = \mathbf{0}$$

- Numerical solution. Matlab/Octave provides the function `fsolve` to find a numerical root
  - Start from an initial guess $\mathbf{x}_0$.
  - Iteratively reduce the residuals until convergence.
- Implementation:
  - `walrasequil.m`: Defines the equilibrium residuals
  - `nlwalras.m`: Calls `fsolve` and computes the equilibrium

Hands-on Exercise 4: Solving the Nonlinear Model

Your Task

Open `nlwalras.m` and `walrasequil.m` in Matlab or Octave

1. **Compute the equilibrium.** Run `nlwalras.m` with $A = 1$; record the equilibrium values of y, c, w, and n
2. **Compare solution methods.** Do you obtain the same results as with `linwalras.m` and `walras.mod`?
3. **Change the initial guess.** Modify the vector `xinit` and solve the model again; does the numerical solution change?
4. **Check the residuals.** Evaluate the four equilibrium equations at the computed solution; are all residuals close to zero?

Hands-on Exercise 5: Nonlinear Model in Dynare

Your Task

Open `walrasnl.mod` and solve the nonlinear Walrasian model using Dynare

1. **Compute the baseline equilibrium.** Run the model with $A = 1$

   ```
   dynare walrasnl.mod
   ```

   Record the equilibrium values of y, c, w, and n

2. **Introduce a productivity shock.** Change the productivity parameter to

   ```
   A = 1.1;
   ```

   and run the model again

3. **Compare the results.** Do the nonlinear and log-linear Dynare models produce the same equilibria?

4. **Examine the model specification.** Why do we set `A = 1.1` here, but `a = log(1.1)` in `walras.mod`?

Walras' Law

- Two markets. Define excess demand for goods and labor

$$z_y = c - y \qquad z_n = n^d - n^s$$

where n^d and n^s denote labor demand and labor supply

- Budget constraint and profits

$$Pc = Wn^s + D \qquad D = Py - Wn^d$$

- Combining both identities:

$$\boxed{P(c-y) + W(n^d - n^s) = 0 \qquad \Rightarrow \qquad Pz_y + Wz_n = 0}$$

- If the labor market clears ($n^d = n^s$), then the goods market must also clear ($c = y$)

Walras' Law

The value of aggregate excess demand is zero. Given the budget constraints, one market-clearing condition is redundant.

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Summary: Static General Equilibrium Models

- Building a general equilibrium model. Specify preferences, technology, individual optimization, and market-clearing conditions
- Determining the equilibrium. Household labor supply and firm labor demand jointly determine employment and the real wage; output and consumption follow from production and goods-market clearing
- Solving the model
  - Exact linear representation in logarithms
  - Nonlinear solution in levels
- Walras' law: Given the budget constraints, one market-clearing condition is redundant

Back to the Opening Puzzle

With logarithmic consumption utility ($\sigma = 1$), a 10% productivity increase raises output, consumption, and the real wage by 10%, but employment remains unchanged.

Literature

 Alogoskoufis, George (2019): Dynamic Macroeconomics, MIT Press, Chapter 2

 Judd, Kenneth L. (1998): Numerical Methods in Economics, MIT Press

 Sydsæter, Knut; Hammond, Peter; Seierstad, Atle; Strøm, Arne (2005): Further Mathematics for Economic Analysis, Prentice Hall

Appendix

Constrained Static Optimization I

(Sydsæter et al. 2005, Chapter 3)

- LAGRANGE-Optimization: minimize or maximize a function of one or more variables subject to constraints
- Function: $f(x_1, x_2, \ldots, x_n)$ with n arguments and m constraints:

$$\begin{aligned} g_1(x_1, x_2, \ldots, x_n) &= \gamma_1 \\ &\ldots \\ g_m(x_1, x_2, \ldots, x_n) &= \gamma_m \end{aligned}$$

- Example:
  1. Function f: utility function $u(c)$
  2. Constraints g: time constraint, capital endowment, budget constraint
  3. Variable: consumption c

Constrained Static Optimization II

(Sydsæter et al. 2005, Chapter 3)

- First step: LAGRANGE function

$$\mathcal{L} = f(x_1, x_2, \ldots, x_n) - \sum_{j=1}^{m} \lambda_j \left(g_j(x_1, x_2, \ldots, x_n) - \gamma_j \right)$$

  where λ_j are called LAGRANGE-multipliers

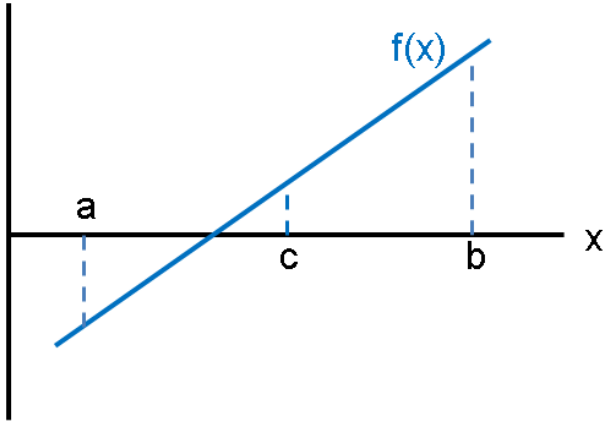
- First-order conditions:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x_i} = \frac{\partial f(x_1, \ldots, x_n)}{\partial x_i} - \sum_{j=1}^{m} \lambda_j \frac{\partial g_j(x_1, \ldots, x_n)}{\partial x_i} &= 0, \qquad i = 1, 2, \ldots, n \\ \frac{\partial \mathcal{L}}{\partial \lambda_j} = g_j(x_1, x_2, \ldots, x_n) - \gamma_j &= 0, \qquad j = 1, 2, \ldots, m. \end{aligned}$$

- $n + m$ equations to be solved for $n + m$ unknowns $x_1, \ldots, x_n, \lambda_1, \ldots, \lambda_m$

Solving Nonlinear Equations: Bisection

(Judd 1998, Figure 5.1)



Bisection: Algorithm

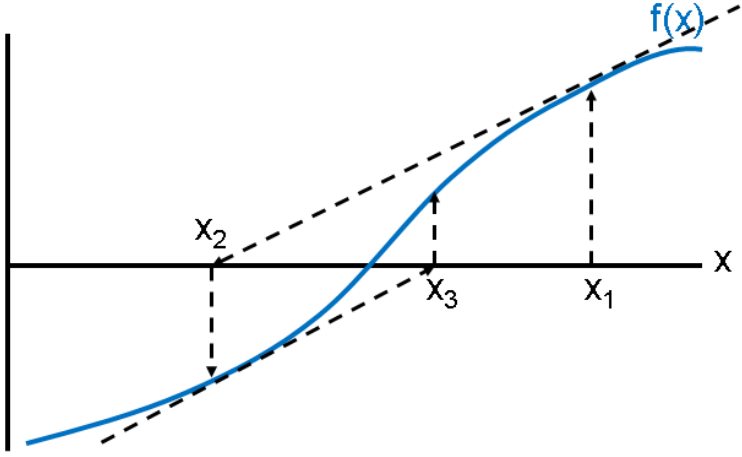
- Find the root (x_0) of $f(x)$: $f(x_0) = 0$
- Initialization
  - Find an interval that includes a root: Find $x^L < x^R$ such that $f(x^L) \cdot f(x^R) < 0$
  - Specify stop criteria ($\epsilon_x = 10^{-5}$, $\epsilon_f = 10^{-5}$, e.g.)
- Step 1: Compute mean: $x^M = (x^L + x^R)/2$
- Step 2: Adjust interval

$$\begin{aligned} x^R &= x^M \quad \text{if } f(x^L) \cdot f(x^M) < 0 \\ x^L &= x^M \quad \text{otherwise} \end{aligned}$$

- Step 3: Check stop criteria
  - Stop: $x_0 = x^M$ if $x^R - x^L \leq \epsilon_x(1 + |x^L| + |x^R|)$ or $|f(x^M)| \leq \epsilon_f$
  - Otherwise go back to step 1

Newton-Method

(Judd 1998, Figure 5.2)



Newton-Method: Algorithm

- Bisection is slow
- Linear approximation $g(x)$ of $f(x)$ at x_k has a root close to the root of $f(x)$

$$g(x) \equiv f'(x_k)(x - x_k) + f(x_k)$$

- Root of $g(x)$:

$$0 = f'(x_k)(x - x_k) + f(x_k)$$

- Iteration:

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

- Stop if

$$|x_k - x_{k+1}| \leq \epsilon_x(1 + |x_k|)$$

- Success only if $|f(x_{k+1})| \leq \epsilon_f$